

Building an AI Agent from Template

Use this document to accompany the fourth video for the "AI Agents for Everyday Professionals" course.

This guide walks you through building your first AI agent workflow using a pre-created template. You'll learn how to set up a "Talk to Your Data" workflow that allows you to have conversations with spreadsheet data using natural language, making data analysis faster and more accessible.

The Problem This Workflow Solves

Analyzing large amounts of data in spreadsheets is tedious and time-consuming. While you can upload spreadsheets to large language models like ChatGPT, Claude, or Gemini, the results are often not as precise, reliable, or scalable. LLMs sometimes hallucinate information or misrepresent data from spreadsheets.

An AI agent workflow provides a more reliable solution. By precisely configuring how your AI interacts with your data, you minimize errors and create a repeatable, scalable process for data analysis.

Step-by-Step Guide

Step 1: Access the N8N Templates

Navigate to your N8N overview page and click **Create Workflow**.

You'll see three options:

- **Manually create your own workflow** (covered in a later video)
- **Prompt it with AI** (currently in beta with unreliable results—skip for now)
- **Start from template** (the easiest option, which we'll use)

Select **Start from template**.

Step 2: Find the Right Template

Click **See more templates** to access the full template library (don't use the featured templates shown initially).

In the search bar, type: "**Talk to your data in sheets with GPT-4**"

Note: The template may reference GPT-4, but you'll configure it to use GPT-4o Mini, which is cheaper and more up-to-date for this simple task.

Step 3: Preview and Import the Template

Click on the template to see a preview of the workflow before committing to it.

In most N8N versions (including trial versions), you can use templates for free. Click **Use for free**.

You have two import options:

- Copy to clipboard and paste into an existing workflow
- **Import into a new workspace** (recommended—select this option)

The template will now be imported and ready to configure.

Step 4: Set Up Your OpenAI API Key

The template includes helpful instructions. Follow them to set up your OpenAI API key credentials.

If you already obtained your OpenAI API key in a previous video:

- Click the dropdown menu in the AI node
- Select your existing OpenAI account credentials
- You only need to add your API key once—it's reusable across workflows

For the **model selection**, choose **GPT-4o Mini**. This is the cheapest, lightest-weight, and most up-to-date model that's perfect for this workflow since it doesn't require intensive computational thinking.

Step 5: Connect Your Google Sheets Account

The workflow needs access to your Google Sheets to read your data.

If you haven't connected Google Sheets to N8N yet:

- Click the Google Sheets node in the workflow
- Follow the prompts to authenticate your Google account
- Grant N8N permission to access your Google Sheets

If you've already connected Google Sheets in a previous workflow, simply select your existing credentials from the dropdown.

Step 6: Select Your Spreadsheet

In the Google Sheets node, select the specific spreadsheet you want to analyze.

Example data structure: The demonstration uses a fake client payment spreadsheet with:

- Client names (company brands)
- Payment amounts (dollar values)
- Payment dates (when money was received)

While the demo spreadsheet has only 50 entries, imagine a real-world scenario with thousands of transactions from hundreds of companies—that's where this workflow truly shines.

Step 7: Activate and Test the Workflow

Once everything is configured, activate your workflow by toggling it on.

The workflow will generate a chat interface where you can ask questions about your data using natural language.

Testing Your Workflow

To ensure your workflow functions correctly, test it with various questions. Here are examples from the demonstration:

Test Question 1: Simple Data Lookup

Question: "How much money did I receive from Apple according to the spreadsheet?"

Expected Result: The AI should correctly identify the payment amount from Apple (in the demo: \$2,450,000)

Verification: Check the spreadsheet to confirm the AI's answer matches the actual data

Test Question 2: Follow-Up Context

Question: "When did I receive this money?"

Expected Result: The AI should understand you're still asking about Apple and provide the payment date (in the demo: January 12th, 2025)

This tests whether the AI maintains conversational context

Test Question 3: Complex Analysis

Question: "Which payments did I receive in the month of May and what were the individual amounts?"

Expected Result: The AI should identify ALL May payments, even if they're not organized chronologically in the spreadsheet

In the demo, it correctly identified four May payments:

- Tesla: \$1,700,000
- Hulu: \$900,000
- Wells Fargo: \$1,600,000

- Oracle: \$800,000

This type of question would take several minutes to answer manually by scrolling through an unorganized spreadsheet

Benefits of This Workflow

- ✓ **Speed:** Instantly answer questions that would take minutes to research manually
- ✓ **Repeatability:** Upload spreadsheets in the same format and continuously query them
- ✓ **Scalability:** Works just as well with 50 rows or 50,000 rows of data
- ✓ **Accessibility:** Ask questions in natural language—no need to write formulas or queries
- ✓ **Accuracy:** More precise than prompting a general LLM because the workflow is purpose-built

Important Security and Accuracy Considerations

■ **Data Security:** When working with sensitive financial data, remove actual names or identifying information from your spreadsheets before uploading them to N8N. Even though N8N is designed to be secure, adding this extra layer of protection is a best practice.

■ **Fact-Checking:** Always verify the AI's conclusions, especially when dealing with financial data. Go back to your spreadsheet and confirm the AI's responses match reality. While this workflow is more accurate than general LLMs, verification is still important for critical decisions.

Common Issues and Solutions

Issue: Chat interface doesn't appear

- Ensure the workflow is activated (toggle switched on)
- Check that all nodes are properly connected
- Verify your OpenAI API key is valid and has available credits

Issue: AI can't find data in spreadsheet

- Confirm the correct Google Sheet is selected
- Verify your Google Sheets connection is active
- Check that your spreadsheet has data in it

Issue: AI provides incorrect answers

- Ensure your spreadsheet data is properly formatted (clear headers, consistent data types)
- Try rephrasing your question more specifically
- Consider whether the model needs more context

Key Takeaways

Congratulations! If you've followed these steps, you've successfully created your first AI agent workflow from a template. You now have a functional tool that can:

- Answer natural language questions about your spreadsheet data
- Maintain conversational context for follow-up questions
- Analyze data more reliably than uploading to a general LLM
- Scale to handle large datasets efficiently
- Save significant time on data analysis tasks

The next step in your journey is building a custom AI agent workflow from scratch, where you'll apply everything you've learned and gain even more control over your automation.